

Education

- 04/2023 - 03/2026** **PhD (Dr. rer. nat)** (Technical University Berlin. Metric Gradient Flows in Measure Spaces: Kernelization and Acceleration. Supervised by Prof. Gabriele Steidl. Referees: Prof. Anna Korba and Prof. José Antonio Carrillo de la Plata. Summa cum laude.)
- 04/2021 – 03/2023** **Mathematics Master's** (Technical University Berlin. Thesis: Wasserstein gradient flows - with an eye towards positive matrix-valued measures. Supervised by Prof. Gabriele Steidl and Dr. Robert Beinert.)
- 10/2017 – 03/2021** **Mathematics Bachelor's** (Technical University Berlin. Thesis: Atomic Norm Minimisation for Superresolution. Supervised by Prof. Gabriele Steidl and Dr. Robert Beinert.)

Positions

- 05/2026 –** **Postdoctoral Researcher** (Department of Mathematics, Technical University of Munich. Chair: Massimo Fornasier; ERC Advanced Grant NEITALG)
- 05/2026 –** **Junior Member** (Munich Center for Machine Learning)
- 05/2026** **One-week research visit (Tim Laux at Heidelberg University)** (Geometry of accelerated dynamics on probability measures in (kernelized) Wasserstein geometries.)
- 11/2025** **One-week research visit (Joaquín Míguez at the University Carlos III Madrid)** (Generative modelling with Bernstein approximations of f -divergences.)
- 08/2024 – 09/2024** **One-month research visit (Wuchen Li at the University of South Carolina)** (Accelerated Stein metric gradient flows with general bilinear kernels on Gaussian families.)
- 10/2025–04/2026** **Phase II Student** (Berlin Mathematical School)
- 04/2023 – 04/2026** **PhD candidate** (Applied Mathematics group at Technical University Berlin. Funding: stipend/BMBF project VI-Screen; TU Berlin teaching appointment from Jan. 2024.)
- 06/2021 – 03/2023** **Student research assistant** (Research on Wasserstein gradient flows, writing lecture notes for “Approximation theory”, rewriting the lecture notes for “Convex Analysis” in the setting of infinite-dimensional spaces, and proofreading manuscripts.)
- 10/2019 – 03/2021** **Tutor** (Department of Mathematics, Technical University Berlin.)

Publications

- 01/2026** **V. Stein, José Manuel de Frutos (equal contribution): Approximating f -Divergences with Rank Statistics** (ICML'26. Proceedings of the 43rd International Conference on Machine Learning)
- 01/2026** **V. Stein, Adwait Datar, Nihat Ay: Well-Posed KL-Regularized Control via Wasserstein and Kalman–Wasserstein KL Divergences** (ICML'26. Proceedings of the 43rd International Conference on Machine Learning)
- 12/2025** **J. Bresch, V. Stein: Interpolating between Optimal Transport and KL regularized Optimal Transport using Rényi Divergences** (Results in Mathematics 81, 23 (2026))

- 07/2025 **R. Duong, V. Stein, R. Beinert, J. Hertrich, G. Steidl: Wasserstein Gradient Flows of MMD Functionals with Distance Kernel and Cauchy Problems on Quantile Functions** (ESAIM: Control, Optimisation and Calculus of Variations.)
- 06/2025 **V. Stein, W. Li: Accelerated Stein Variational Gradient Flow** (In: Nielsen, F., Barbaresco, F. (eds) Geometric Science of Information. GSI 2025. Lecture Notes in Computer Science, vol 16034. Springer, Cham, pp. 80–90.)
- 06/2025 **R. Duong, N. Rux, V. Stein, G. Steidl: Wasserstein Gradient Flows of MMD Functionals with Distance Kernels under Sobolev Regularization** (Philosophical Transactions of the Royal Society A, vol. 383, issue 0243 "Partial differential equations in data science".)
- 04/2025 **V. Stein, S. Neumayer, N. Rux, G. Steidl: Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces** (Analysis and Applications Vol. 24, No. 01, pp. 21-65 (2026).)

Preprints

- 03/2026 **SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds** (with Wuchen Li, University of South Carolina, and Gabriele Steidl, Technical University Berlin.)
- 09/2025 **Towards understanding Accelerated Stein Variational Gradient Flow - Analysis of Generalized Bilinear Kernels for Gaussian target distributions** (with Wuchen Li, University of South Carolina.)

Conference and Workshop Talks

- 11/2026 **SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds** (SIAM Conference on Mathematics of Data Science in Utah, USA.)
- 11/2026 **Wasserstein gradient flows of MMD-regularized f -divergences** (SIAM Conference on Imaging Science in Utah, USA.)
- 09/2026 **Wasserstein gradient flows of MMD-regularized f -divergences** (BMS – BGSMATH Junior Meeting 2026 in Barcelona, Spain.)
- 07/2026 **Accelerated Stein Variational Gradient Flow** (Minisymposium on "New trends in geometric machine and deep learning" at CMAM 2026: Computational Methods in Applied Mathematics in Vienna, Austria.)
- 06/2026 **SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds** (Learning and Optimization in Luminy 2026)
- 03/2026 **SympFormer: Accelerated attention blocks via Inertial Dynamics on Density Manifolds** (MFO Workshop 2613 - Flows on Measure Spaces and Applications in Machine Learning in Oberwolfach, Germany.)
- 02/2026 **What does Adam have to do with symplectic manifolds? The geometry behind momentum methods in machine learning** (14th BMS Student Conference, Berlin.)
- 10/2025 **Accelerated Stein Variational Gradient Flow** (GSI'25: Geometric Structures of Statistical & Quantum Physics, Information Geometry, and Machine Learning (Saint-Malo, France), fully funded by the DAAD.)
- 09/2025 **Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces** (MML'25: Conference on Mathematics of Machine Learning 2025 in Hamburg, Germany.)

Seminar and Colloquium Talks

11/2025	Accelerated Stein Variational Gradient Flow (At the colloquium of the Signal Processing Department of University Carlos III Madrid.)
04/2025	Accelerated Stein Variational Gradient Flow (Stan Osher's UCLA level set seminar (online).)
09/2024	Interpolating between Optimal Transport and KL regularized Optimal Transport using Rényi Divergences. (University of South Carolina Mathematics Graduate Colloquium)
08/2024	Wasserstein Gradient Flows of MMD Functionals with Distance Kernel and Cauchy Problems on Quantile Functions. (Joint Applied and Computational Mathematics (Changhui Tan & Siming He) and RTG data science seminar (Wuchen Li), University of South Carolina.)
08/2024	Wasserstein Gradient Flows for Moreau Envelopes of f-Divergences in Reproducing Kernel Hilbert Spaces (Stan Osher's UCLA level set seminar (online).)

Poster Presentations

07/2026	FoCM 2026, Vienna. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces (Computational Optimal Transport Session), Accelerated Stein Variational Gradient Flow (Foundations of Data Science and Machine Learning session))
07/2026	ICML 2026, Seoul. ("Approximating f -Divergences with Rank Statistics" & "Well-Posed KL-Regularized Control via Wasserstein and Kalman-Wasserstein KL Divergences".)
03/2026	TUM's research retreat "Advanced Mathematical Methods in Data Analysis and Signal Processing", Tutzing, Germany. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces (best poster prize))
02/2026	Gradient structures workshop in Augsburg, Germany. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces)
08/2025	Mathematical and Scientific Machine Learning, Naples, Italy. (Accelerated Stein Variational Gradient Flow)
10/2024	SIGMA (Signal - Image - Geometry - Modelling - Approximation) Workshop at the CIRM in France. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces)
06/2024	LOL: Learning and Optimization in Luminy, CIRM, France. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces)
03/2024	Workshop on Optimal transport from theory to applications, Berlin, Germany. (Wasserstein Gradient Flows for Moreau Envelopes of f -Divergences in Reproducing Kernel Hilbert Spaces)

Teaching at TU Berlin

Winter 2025/26	Harmonic Analysis (Lecture assistant)
Winter 2025/26	Numerical Mathematics I (Tutor)
Summer 2025	Mathematical Physics II - Statistical Mechanics (Lecture assistant)
Summer 2025	Probability Theory I (Tutor)
Winter 2024/25	Analysis II for Mathematicians (Tutor)

Winter 2024/25	Harmonic Analysis (Lecture assistant)
Summer 2024	Convex Analysis (Lecture assistant)
01/2024 - 02/2024	Numerical Mathematics I (Lecture assistant)

Supervised Thesis

03/2025	Roxane Leitheiser, Technical University Berlin (Wasserstein Gradient Flows of the MMD with Riesz Kernels on the Real Line. Bachelor's thesis. First supervisor: Gabriele Steidl.)
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Service

- Reviewer: JOTA, IPI, TMLR, IEEE/CAA Journal of Automatica Sinica, IEEE TNNLS, NeurIPS 2024 workshop.
- Reporter: Oberwolfach Report 26/13, Flows on Measure Spaces and Applications in Machine Learning.

Awards

Best Bachelor's Thesis Talk at the 17th annual Dies Mathematicus (TU Berlin, 2022).
Best poster prize at the TUM's Research retreat "Advanced Mathematical Methods in Data Analysis and Signal Processing".

Technical Skills

Python, PyTorch, MATLAB, $\text{\LaTeX}$; experience with HPC clusters.

Volunteer Work

In the school year 2022/23, I have been tutoring around 14 seventh-graders in weekly sessions, discussing mathematical puzzles and questions from the German Mathematical Olympiad. I have also served as a corrector at the team competition at the Tag der Mathematik 2022 (Mathematics Day), organized by the three Berlin universities, where sixty-nine teams of high schoolers participated.

Languages

German native; English fluent; French A2.

References

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